

FAIRCHILD AFB, WA, USA

WMO: 727855

Lat: 47.633N

Lon: 117.650W

Elev: 2461

StdP: 13.44

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 24114

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	6.5	11.7	-1.4	5.6	9.4	4.1	7.4	15.8	28.1	39.5	25.2	35.2	7.1	30	0.602

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	27.4	92.6	62.0	89.6	61.2	86.0	60.5	64.7	84.5	63.2	83.4	61.8	81.5	7.8	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
59.1	81.9	67.3	56.5	74.6	65.5	54.4	68.9	64.9	30.9	84.3	29.7	83.3	28.6	81.6	72.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.4	20.6	18.3	DB	-0.9	98.0	8.2	3.1	-6.8	100.3	-11.6	102.1	-16.3	103.8	-22.3	106.1	(4)
(5)			WB	0.0	67.5	8.9	2.2	-6.4	69.1	-11.6	70.4	-16.6	71.6	-23.0	73.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.8	28.7	31.9	39.3	46.1	54.5	60.9	70.3	69.1	60.1	46.8	36.3	28.5
	DBStd	16.20	8.20	7.82	6.60	6.36	7.21	7.04	6.49	6.47	7.33	7.05	7.46	7.90
	HDD50	2923	661	508	332	151	37	3	0	0	9	144	412	666
	HDD65	6707	1126	928	795	566	333	162	28	35	179	563	861	1131
	CDD50	2124	0	0	2	36	176	331	628	591	313	46	1	0
	CDD65	432	0	0	0	0	7	39	191	161	33	1	0	0
	CDH74	5936	0	0	0	6	141	584	2580	2093	519	13	0	0
	CDH80	2408	0	0	0	0	22	184	1154	893	155	0	0	0
Wind	WSAvg	8.5	8.3	8.9	9.8	9.7	8.9	8.5	7.7	7.6	7.5	8.2	8.6	8.5
Precipitation	PrecAvg	15.6	1.9	1.4	1.5	1.2	1.5	1.1	0.6	0.6	0.7	1.0	2.1	2.2
	PrecMax	25.2	4.5	3.3	3.8	2.6	3.7	2.8	2.3	1.9	1.9	3.3	4.8	4.9
	PrecMin	10.5	0.4	0.0	0.2	0.1	0.2	0.1	0.0	0.0	0.0	0.0	0.2	0.6
	PrecStd	2.9	0.9	0.7	0.7	0.7	0.8	0.6	0.6	0.6	0.5	0.8	1.1	1.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.9	52.7	63.0	74.5	82.8	91.4	97.5	96.2	90.3	75.2	57.2	50.2
		MCWB	45.1	44.7	48.4	55.3	59.1	62.7	62.9	62.3	60.5	56.6	50.2	47.0
	2%	DB	43.5	48.1	57.6	68.3	78.1	85.1	93.2	92.0	84.2	68.0	52.7	44.9
		MCWB	41.0	43.6	46.8	52.3	57.7	60.8	62.5	61.6	58.7	52.7	48.4	42.8
	5%	DB	40.9	44.8	54.2	63.3	73.5	80.7	90.2	89.0	80.2	63.2	49.7	41.1
		MCWB	39.1	40.8	45.1	49.5	55.8	59.1	61.4	61.0	57.6	50.8	45.8	39.5
	10%	DB	38.3	41.8	50.3	58.9	69.4	76.5	86.8	85.5	75.5	58.7	46.4	38.5
		MCWB	37.0	38.8	43.2	47.2	53.8	57.4	60.9	60.4	56.0	48.5	43.3	37.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.6	47.2	51.6	57.3	61.6	65.7	68.3	65.9	62.9	58.6	52.8	47.1
		MCDB	46.2	49.8	57.1	70.3	73.5	85.1	84.7	85.7	83.2	69.2	54.6	49.1
	2%	WB	41.7	44.4	48.6	53.6	59.3	62.6	65.5	64.0	60.3	55.2	49.4	43.0
		MCDB	43.1	47.0	55.0	64.1	73.9	80.2	84.3	84.9	79.1	64.1	52.2	44.6
	5%	WB	39.2	41.6	46.3	50.9	57.3	60.6	63.8	62.5	58.7	52.5	46.4	39.8
		MCDB	40.5	43.8	51.8	60.9	70.5	76.9	84.2	83.4	76.5	60.5	48.4	40.8
	10%	WB	37.1	39.1	44.4	48.6	55.1	58.8	62.2	61.1	56.9	49.9	44.2	37.3
		MCDB	38.1	41.3	49.4	57.3	66.8	73.5	82.5	81.8	73.2	57.2	46.1	38.3
Mean Daily Temperature Range	5% DB	MDBR	9.4	12.6	17.0	19.8	21.7	22.9	27.4	26.7	24.9	19.4	12.5	9.2
		MCDBR	11.0	15.6	22.6	26.3	27.7	28.6	31.3	31.3	30.9	25.6	15.8	11.0
		MCWBR	9.0	11.0	13.5	13.4	11.9	10.7	10.7	11.4	12.5	13.1	11.1	9.5
	5% WB	MCDBR	10.7	13.9	19.4	23.4	24.8	26.1	27.9	28.2	27.5	22.2	13.8	10.5
MCWBR		9.1	10.5	12.8	13.0	11.3	10.6	10.2	11.0	12.1	12.5	10.7	9.5	
Clear-Sky Solar Irradiance	taub	0.270	0.278	0.290	0.310	0.324	0.326	0.325	0.333	0.311	0.298	0.278	0.266	
	taud	2.503	2.544	2.570	2.492	2.474	2.496	2.500	2.458	2.512	2.547	2.559	2.507	
	Ebn at Noon	265	287	299	300	298	297	296	289	287	275	260	252	
	Edh at Noon	19	23	26	31	33	33	32	32	28	23	18	17	
All-Sky Solar Radiation	RadAvg	331	634	978	1461	1865	2011	2344	1948	1394	830	397	262	
	RadStd	42	85	108	133	145	168	125	126	125	91	44	35	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (19 neighbors)		N/A	N/A	N/A	N/A	-0.72	N/A	N/A	N/A	+166	+81	

Nomenclature: See separate page